ONENET SOLUTIONS · DEPLOYMENT GUIDE

HR & Payroll Suite
New Client Server Setup

How to install a ready-to-use HR & Payroll system for a new client: on their own Ubuntu server or yours, with or without aaPanel, in the Full or the Lite edition.

1 · Prepare
Server, DNS, ports, client details

2 · Build
One .run file on your PC

3 · Install
sudo bash … about 5 minutes

4 · Hand over
Sign-in, edition, backups

Bundle
2026.10.19-71f717a-dirty

Editions
Full · Lite

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1 · What the installer does

You build **one file** per release, `hrpayroll-setup-2026.10.19-71f717a-dirty.run`. Run it as root on a fresh or existing Ubuntu server. A guided, menu-driven wizard asks a few questions, then the installer does everything else without further input.

PART	WHAT HAPPENS
Node.js	A private, checksum-verified Node.js 24 runtime in `/opt/hrpayroll/node`. It never touches aaPanel's Node manager or other websites.
PostgreSQL	Uses aaPanel's PostgreSQL or the system's. If neither exists, it installs PostgreSQL 16. It then creates a database and user for this client with a random password.
Application	Installed as a versioned release with a `current` link, running as its own Linux user. All database migrations are applied.
Edition	Sets **Full** or **Lite**, using the same code (and audit-log entry) as the OneNet edition console.
Service	A hardened systemd service `hrpayroll-<id>`. It starts on boot and restarts on failure, and it listens only on 127.0.0.1.
Website & HTTPS	An nginx site for the client's domain, a free Let's Encrypt certificate with automatic renewal, and HTTP→HTTPS redirection.
First admin	The client's Super Admin account and company are created, so they can sign in immediately.
Backups	An encrypted database backup every night at 02:15 (the last 14 are kept), plus one straight after installation.

> **One bundle fits every client.**
> The web app is built to call the server that served it, so the same `.run` file works for any domain. Nothing is customised per client at build time.

2 · Before you start

Server

- [ ] Ubuntu **24.04** or 22.04 LTS, 64-bit (x86-64 or ARM64)
- [ ] **2 GB RAM** or more (1 GB works for Lite with few users)
- [ ] At least **2 GB free** disk on `/var`
- [ ] SSH or aaPanel terminal access as **root** / sudo
- [ ] Internet access (packages, Node.js, Let's Encrypt)

Network

- [ ] DNS **A record**: `hr.client.com` → server's public IP
- [ ] Ports **80** and **443** open in aaPanel *Security* → *Firewall*
- [ ] …and in the cloud provider's firewall / security group
- [ ] Nothing except nginx using port 80
- [ ] The domain **not** already added under aaPanel *Website*

Collect from the client

ITEM	EXAMPLE	NOTES
Domain	`hr.acme-traders.com`	DNS must already point at the server
Company name	Acme Traders (Pvt) Ltd	Shown in the app, titles, MFA
Edition	Full or Lite	See section 3
Super Admin name & email	Ali Khan, admin@acme-traders.com	Their first sign-in
Admin password	12+ characters	Or let the installer generate one
Email for SSL notices	it@acme-traders.com	Optional
SMTP (payslips, sign-in codes)	smtp.gmail.com : 587 + app password	Optional; can be added later

A printable intake form is in the appendix.

3 · Choosing the edition: Full or Lite

Full is the complete Pakistan payroll-compliance system. **Lite** is for shops and small businesses with about 10–15 staff, who only need attendance, leave, salaries, advances and payslips. Lite does not ask for FBR, EOBI or social-security data, and one owner can run payroll alone.

FEATURE	FULL	LITE
Statutory deductions		
FBR income tax withholding Tax slabs, NTN, salary tax, CPR deposits, annual tax certificates	✓ On	— Off
EOBI contributions Employee and employer EOBI, registration numbers, returns	✓ On	— Off

Provincial social security (PESSI / SESSI) Employer contribution and returns	✓ On	— Off
Provident fund PF deduction, employer match, PF ledger	✓ On	— Off
Gratuity and final settlement Gratuity provision and final-settlement runs	✓ On	— Off
Payroll controls		
Bank payroll Off: cash salaries allowed, bank details optional	✓ On	— Off
Maker-checker approvals Off: one owner can prepare, approve and pay	✓ On	— Off
Employee email required Off: email is optional	✓ On	— Off
Modules		
Recruitment Job postings and applicants	✓ On	— Off
Performance reviews KPI scoring and appraisals	✓ On	— Off
Asset management Assets issued to staff, exit clearance	✓ On	— Off
Loans and advances Recovered through payroll	✓ On	✓ On
Salary revisions Increments and promotions with arrears	✓ On	✓ On
Parallel-run reconciliation Compare against a legacy payroll file	✓ On	— Off
Live payroll readiness checklist Statutory, MFA and approvals checklist	✓ On	— Off

The edition can be changed at any time.

Switching never deletes or changes customer data. Moving back to Full restores every rule. Use `sudo hrpayrollctl edition <id> lite|full` on the server, or the OneNet vendor console on the sign-in screen (section 10).

4 · Step 1 — Build the setup bundle (your PC)

In the project folder on your development PC (Windows or Linux), run:

```
PS> cd D:\GIT-HUB\HR-Payroll-Management-System
PS> npm run server:bundle            # optional: -- --version 1.5.0
```

About 15 seconds later `deploy\installer\dist\` contains:

FILE	USE
`hrpayroll-setup-<version>.run`	Installs a new client, and updates existing ones
`hrpayroll-app-<version>.tar.gz`	The app only, as an alternative update package
`SHA256SUMS`	Checksums for verifying a copy
`New-Client-Server-Setup-Guide.pdf`	This guide

> **Build from committed, tested code.**
> The version name ends in `-dirty` if there are uncommitted changes. Release bundles should come from a clean, committed state. The build also refuses to finish if a domain from your `.env` files ended up inside the web app.

5 · Step 2 — Copy it to the client's server

Use any one of these:

- **aaPanel** → **Files**: upload the `.run` file to `/root`.
- **WinSCP / FileZilla**: SFTP to the server, **binary** mode, into `/root`.
- **Command line** from your PC:

```
PS> scp deploy\installer\dist\hrpayroll-setup-2026.10.19-71f717a-dirty.run root@SERVER-IP:/root/
```

Optionally, check the file arrived intact:

```
# sha256sum /root/hrpayroll-setup-2026.10.19-71f717a-dirty.run      # must match SHA256SUMS
# bash /root/hrpayroll-setup-2026.10.19-71f717a-dirty.run --info
```

6 · Step 3 — Run the setup wizard

Open an SSH session (or aaPanel → Terminal) and run:

```
# sudo bash /root/hrpayroll-setup-2026.10.19-71f717a-dirty.run
```

Use the arrow keys, **Tab** and **Enter** to move through the blue setup screens:

1. **Welcome**: shows what was detected on this server (aaPanel, PostgreSQL, nginx).
2. **Client domain**: e.g. `hr.acme-traders.com`.
3. **Instance id**: a short name for this client, such as `acme`. It is used for the service, database and folders. Accept the suggestion unless it clashes.
4. **Company**: the name shown in the app.
5. **Edition**: choose **Full** or **Lite**.
6. **Administrator**: full name, sign-in email and password (12+ characters). Leave the password blank to generate one; it is shown once at the end.
7. **HTTPS**: **Yes** gets a Let's Encrypt certificate. You can give an email for expiry notices.
8. **Email delivery**: optionally enter the SMTP settings now.
9. **Payroll tax** (Full only): load the Pakistan FBR tax slabs. Lite skips this.
10. **PostgreSQL password**: only asked if the installer cannot sign in to aaPanel's PostgreSQL on its own. Find it in aaPanel → Databases → PostgreSQL → root password.
11. **Ready to install**: check the summary and press **Yes**.

Each step then shows a tick as it completes. A typical install takes 3–5 minutes, most of it downloading packages.

7 · Step 4 — Finish screen and handover

```
✓ HR & Payroll Suite is live for Acme Traders (Pvt) Ltd

  Open:        https://hr.acme-traders.com
  Sign in:     admin@acme-traders.com
  Password:    (the one you chose)
  Edition:     Lite
  Manage:      hrpayrollctl status acme | logs acme | backup acme | update acme <bundle.run>
  Summary:     /etc/hrpayroll/acme/install-summary.txt
```

Handover checklist

- [ ] **Save the two keys off the server.** Copy `BACKUP_ENCRYPTION_KEY` and `MFA_ENCRYPTION_KEY` from `/etc/hrpayroll/<id>/app.env` into your password manager. **Without the backup key, the backups cannot be restored.**
- [ ] Open the website, sign in as the Super Admin, and complete **Company Setup** (legal name, logo, branches). The installer creates a "Head Office" branch.
- [ ] Check the edition the client sees in the menu: Lite hides statutory pages and modules.
- [ ] Enrol authenticator MFA for every admin and payroll user. Then run `hrpayrollctl edit <id>`, set `MFA_ENFORCE_PRIVILEGED=true`, and save.
- [ ] **Windows kiosk**: install it on the client's attendance PC and set its server address to the client's URL (`PORTAL_URL`). By default the kiosk connects to OneNet's own server.
- [ ] **Mobile app**: sign in with the client's URL.
- [ ] Optionally, copy `/var/backups/hrpayroll/<id>` to off-site storage (aaPanel cron, rclone or similar).
- [ ] Delete any client config file that contains passwords (section 8).

8 · Unattended installs with a client config file

For repeat rollouts, keep one small file per client and install with no questions asked:

```
# bash hrpayroll-setup-2026.10.19-71f717a-dirty.run --extract-only /tmp/hrp && cp
/tmp/hrp/installer/client.conf.example acme.conf
# nano acme.conf
# sudo bash hrpayroll-setup-2026.10.19-71f717a-dirty.run --config acme.conf --yes
```

```
# acme.conf
DOMAIN=hr.acme-traders.com
COMPANY_NAME="Acme Traders (Pvt) Ltd"
EDITION=lite
ADMIN_NAME="Ali Khan"
ADMIN_EMAIL=admin@acme-traders.com
ADMIN_PASSWORD=             # blank = generate and print once
SSL=yes
SSL_EMAIL=it@acme-traders.com
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=payroll@acme-traders.com
SMTP_PASS=app-password-here
```

Every setting can also be given as an option, e.g. `--domain`, `--company`, `--edition lite`, `--admin-email`, `--no-ssl` or `--no-vendor-console`. Run the bundle with `--help` for the full list.

9 · Running several clients on one server

Run the same bundle again with another domain. Each client is fully separate:

PER CLIENT	LOCATION (INSTANCE ID `ACME`)
Application	`/www/wwwroot/acme` (aaPanel) or `/opt/hrpayroll/apps/acme`
Settings & secrets	`/etc/hrpayroll/acme/app.env`
Database / user	`hrp_acme` / `hrp_acme`
Employee documents	`/var/lib/hrpayroll/acme/documents`
Backups	`/var/backups/hrpayroll/acme`
Service · port	`hrpayroll-acme` · 8080, 8081, … assigned automatically
Logs	`/var/log/hrpayroll/acme-*.log` and `hrpayrollctl logs acme`

10 · Day-to-day management with hrpayrollctl

COMMAND (RUN WITH SUDO)	WHAT IT DOES
`hrpayrollctl list`	All clients on this server, with state and version
`hrpayrollctl status acme`	Service, health, version, edition, certificate expiry, last backup
`hrpayrollctl logs acme -f`	Live application log
`hrpayrollctl restart acme`	Restart and wait until healthy (also `start`, `stop`)
`hrpayrollctl edition acme`	Show the current edition
`hrpayrollctl edition acme lite`	Switch to Lite (or `full`). Users see it within seconds; no data changes.
`hrpayrollctl edit acme`	Edit settings (SMTP, SMS/WhatsApp, MFA enforcement…) and restart
`hrpayrollctl backup acme`	Encrypted backup now
`hrpayrollctl restore acme <file.dump.enc>`	Replace the database with a backup (asks you to confirm)
`hrpayrollctl ssl acme`	Issue or repair HTTPS (e.g. once DNS is fixed)
`hrpayrollctl rollback acme`	Go back to the previous version
`hrpayrollctl uninstall acme`	Remove the service and website, keeping the data
`hrpayrollctl uninstall acme --purge`	Also delete the database, files and documents (backups are kept unless `--delete-backups`)

Switching the edition from the browser (vendor console)

On the client's sign-in screen, enter login `onenet` and the password `OneNet` followed by today's date as `DDMonYYYY` on Pakistan time, e.g. `OneNet19Oct2026`. This opens the edition console, which can change **only** the edition. The change is recorded in the audit log.

11 · Updating clients to a new version

1. Build the new bundle (Step 1) and copy it to the server (Step 2).
2. Run, for each client on that server:

```
# sudo hrpayrollctl update acme /root/hrpayroll-setup-<new-version>.run
```

3. The update takes a backup, unpacks the new version, applies migrations, switches over and runs a health check. If the new version is unhealthy, it **switches back automatically**.

The client's edition, settings, documents and data are kept. Only the program changes.

> **Rolling back does not undo database migrations.**
> If you roll back after an update that changed the database structure, the pre-update backup is in the client's backup folder.

12 · aaPanel notes

- The website is a configuration file at `/www/server/panel/vhost/nginx/hrpayroll-<id>.conf`. It does **not** appear under aaPanel *Website → Sites*, so **don't add the same domain there**. The installer stops if the domain is already used by an aaPanel site.
- The app runs as a systemd service, **not** under aaPanel's Node Project Manager. Don't add it there as well, or two copies will fight over the port.
- aaPanel's own firewall must allow 80 and 443. The installer opens them in `ufw` if it is active.

13 · Troubleshooting

SYMPTOM	WHAT TO DO
"HTTPS could not be issued yet"	The site works over HTTP. Fix DNS so the domain points at this server, and make sure port 80 is open, then run `hrpayrollctl ssl <id>`.
A step shows a red ✕	The last lines of the log are printed, and the full log is in `/var/log/hrpayroll/<id>-install.log`. Fix the cause, run `hrpayrollctl uninstall <id> --purge`, then run the bundle again.
"This setup file is damaged"	The file was changed in transfer, e.g. FTP in text mode. Copy it again in binary mode and compare with `SHA256SUMS`.
"…is already configured in …conf"	The domain is already used by an aaPanel website. Delete that site in aaPanel, then re-run.
"Cannot reach PostgreSQL as superuser"	Give the aaPanel PostgreSQL root password when asked, or pass `--pg-password`.
"Port 80 is used by something other than nginx"	Stop Apache (or whatever else is using port 80), or use a server running nginx.
Site down after editing settings	Run `hrpayrollctl logs <id>`. The app refuses to start with `DEBUG=true` or an empty `APP_URL`.
Kiosk says it cannot reach the server	Its server address still points at OneNet's server. Set the client's URL (`PORTAL_URL`).
Client sees the wrong menus	Check the edition with `hrpayrollctl edition <id>`, switch it, and reload the page.

14 · Security notes

- Each client gets its own Linux user, database user and randomly generated secrets. The application listens only on 127.0.0.1, behind nginx and HTTPS.
- `app.env` and `install-summary.txt` can be read only by root and the app's own user. Never send them by email or chat.
- Treat client config files (`*.conf`) like passwords, and delete them after installing.

> **The vendor console password follows a predictable pattern.**
> Anyone who learns the "OneNet + today's date" pattern can switch *any* client between Full and Lite. Switching to Lite turns off maker-checker approvals, so a single user could then approve and pay payroll. For clients where that matters:
>
> - install with `--no-vendor-console` and switch editions with `hrpayrollctl edition` on the server; or
> - set a private `VENDOR_PASSWORD_PREFIX` per client with `hrpayrollctl edit <id>`.

Appendix — Client intake form

Fill one in per client before installing. Keep it with the client's file; don't write passwords on it.

CLIENT
Company legal name
Contact person & phone
Number of employees
Edition (Full / Lite)
SERVER
Hosting provider · server IP
Ubuntu version · aaPanel (yes/no)
Domain (DNS A record set?)
Ports 80/443 open (panel + cloud)
ACCOUNTS
Super Admin name
Super Admin email
SSL notice email
SMTP host · port · user
AFTER INSTALLATION
Instance id · port
Bundle version installed
Keys saved to password manager (date / by)
Kiosk installed & pointed at client URL
MFA enrolled · enforcement on
Handed over to client (date / signature)